

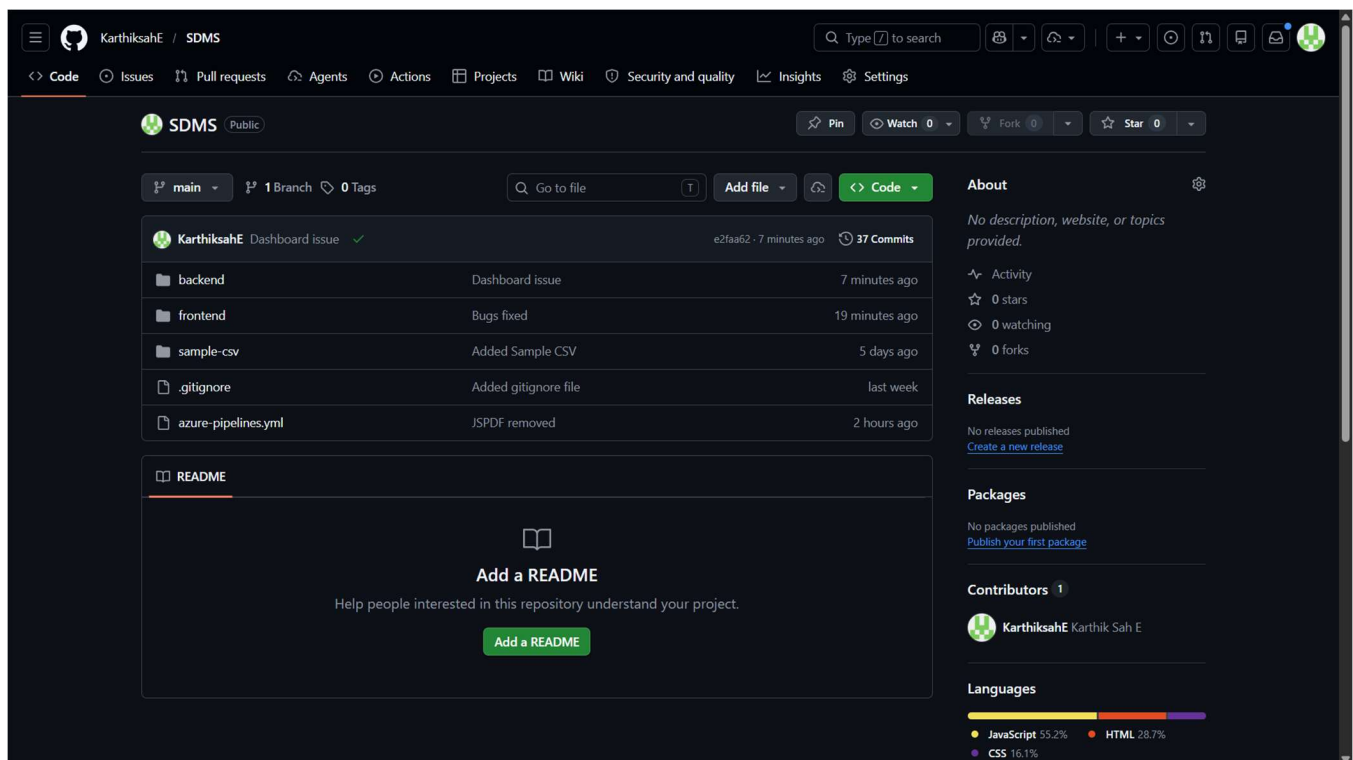
EXP NO: 10

GITHUB: PROJECT STRUCTURE & CI/CD PIPELINE

Aim:

To design and maintain a well-structured GitHub repository with proper naming conventions, and to implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline using Azure DevOps for automating the build, testing, and deployment of the Student Data Management System, ensuring efficient development and improved software quality.

GitHub Project Structure



CI/CD PIPELINE IN AZURE DEVOPS

PROCEDURE:

1. Sign in to Azure DevOps
 - Navigate to: <https://dev.azure.com>
 - Select organization and project
2. Connect Repository
 - Link the GitHub repository to Azure DevOps
 - Select the SDMS repository
3. Create Pipeline
 - Go to Pipelines → Create Pipeline

- Choose GitHub as source
- Select repository

4. Configure Pipeline (YAML)

A YAML file (azure-pipelines.yml) is used for automation.

5. Define CI (Build Stage)

The following steps are executed:

- Install Node.js
- Install dependencies (npm install)
- Run tests (npm test)
- Prepare application for deployment

6. Copy Frontend to Backend

Frontend files are copied into backend:

xcopy frontend backend\public /E /I /Y

7. Archive Application

- Application is compressed into a ZIP file
- Stored as build artifact

8. Publish Artifacts

- Artifacts are published for deployment stage

9. Run Pipeline

- Click Run Pipeline
- Azure executes all steps automatically

CONTINUOUS DEPLOYMENT (CD)

10. Create Release Pipeline

- Add artifact from build stage
- Create deployment stages

11. Deploy to Azure App Service

- Use Azure Web App task
- Configure:
 - Subscription
 - App name
 - Deployment package

12. Enable Triggers

- Enable automatic deployment after build
- Configure approvals if needed

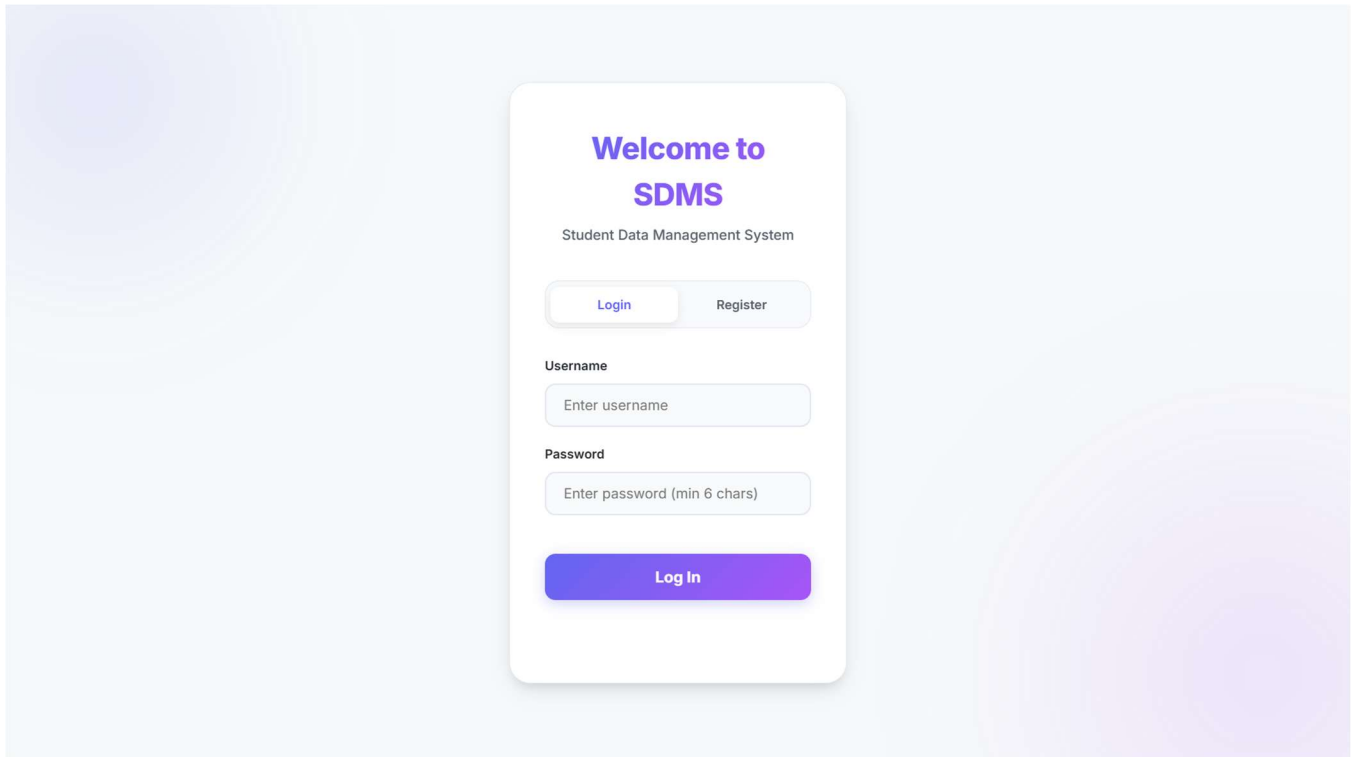
13. Monitor Pipeline

- View logs for each stage
- Debug errors if any

14. Maintain Pipeline

- Update YAML when project changes

- Ensure secure and efficient builds



The image shows a login form for the Student Data Management System (SDMS). The form is centered on a light blue background with abstract circular patterns. It has a white card-like appearance with rounded corners and a subtle shadow. The title 'Welcome to SDMS' is in bold purple, with 'SDMS' being larger. Below it, 'Student Data Management System' is in a smaller, grey font. There are two buttons at the top: 'Login' (active, purple) and 'Register' (inactive, light blue). The 'Username' section has a text input with the placeholder 'Enter username'. The 'Password' section has a text input with the placeholder 'Enter password (min 6 chars)'. At the bottom is a large purple 'Log In' button.

**Welcome to
SDMS**

Student Data Management System

Login Register

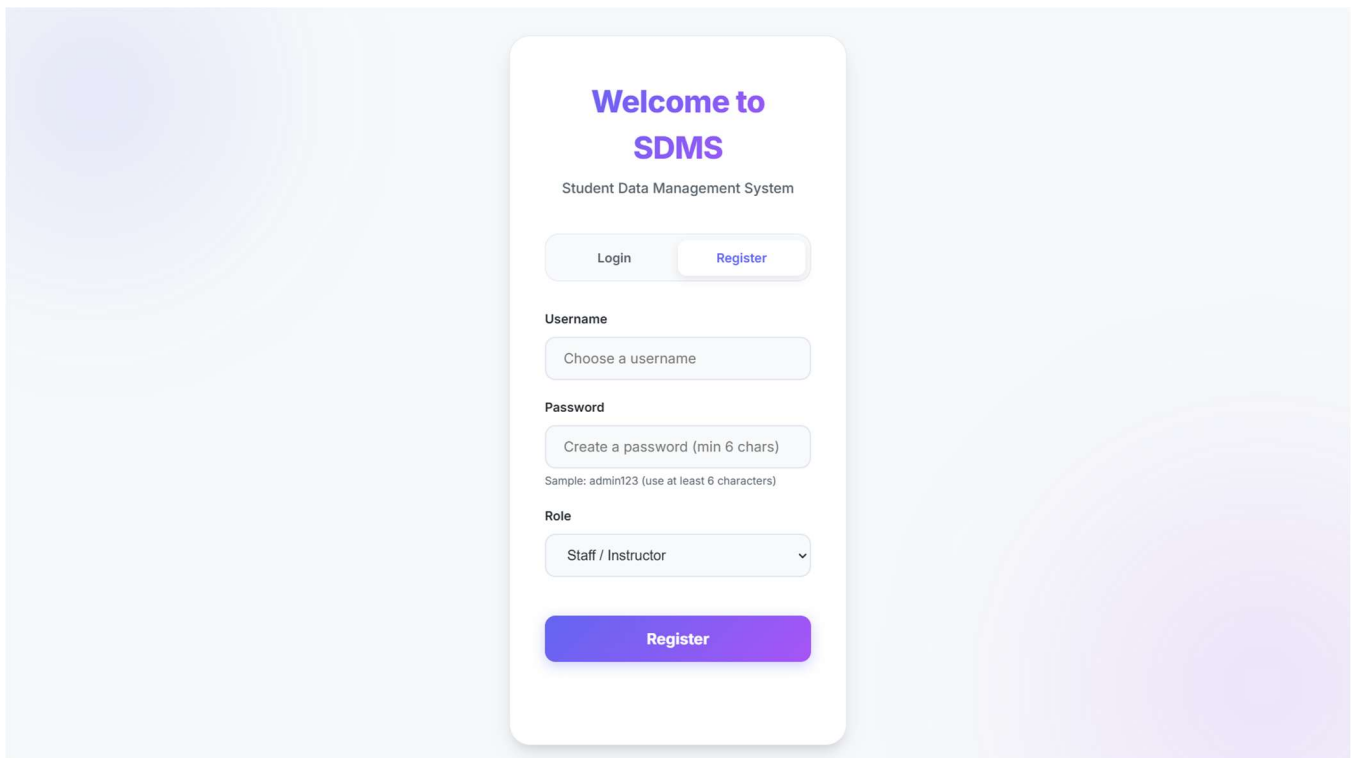
Username

Enter username

Password

Enter password (min 6 chars)

Log In



The image shows a registration form for the Student Data Management System (SDMS). It follows the same design as the login form, with a white card on a light blue background. The title 'Welcome to SDMS' is in bold purple. Below it, 'Student Data Management System' is in a smaller, grey font. The buttons at the top are 'Login' (inactive, light blue) and 'Register' (active, purple). The 'Username' section has a text input with the placeholder 'Choose a username'. The 'Password' section has a text input with the placeholder 'Create a password (min 6 chars)' and a sample text 'Sample: admin123 (use at least 6 characters)'. The 'Role' section has a dropdown menu with 'Staff / Instructor' selected. At the bottom is a large purple 'Register' button.

**Welcome to
SDMS**

Student Data Management System

Login Register

Username

Choose a username

Password

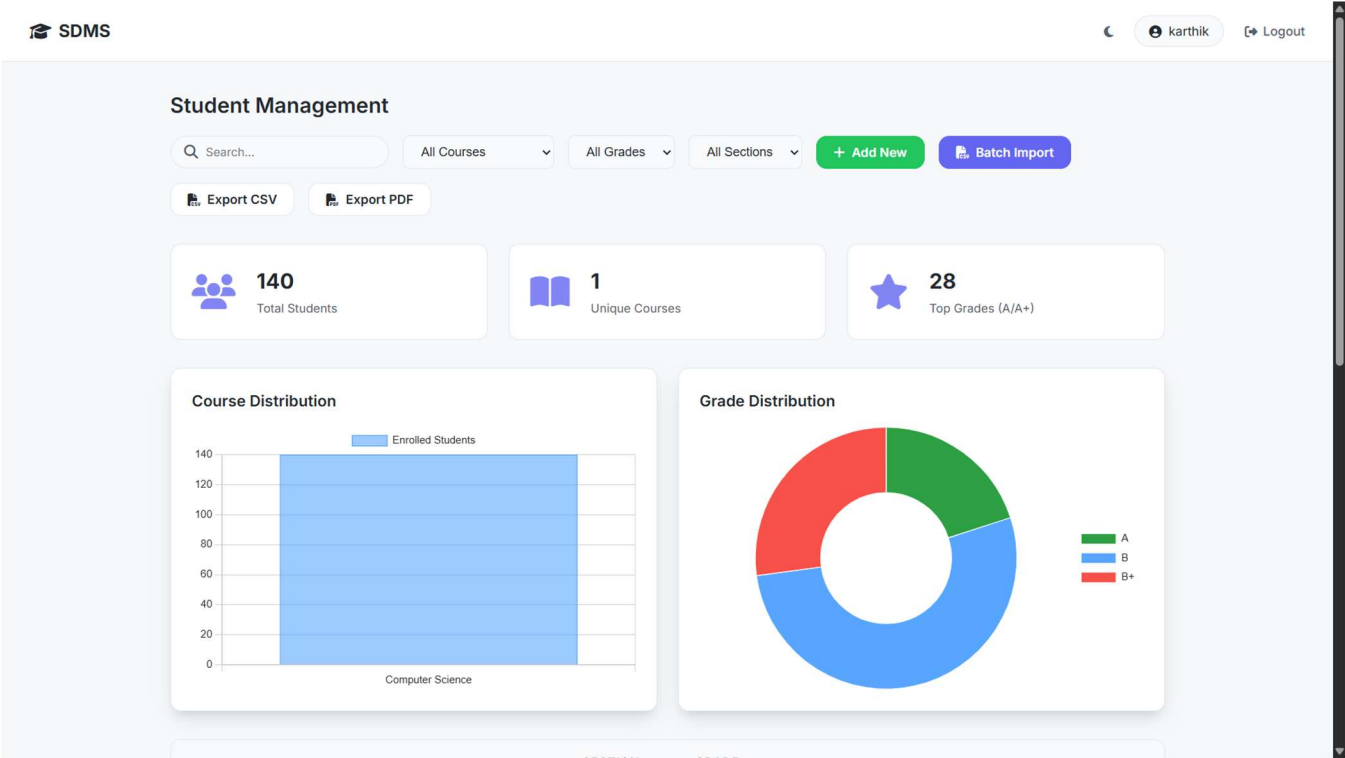
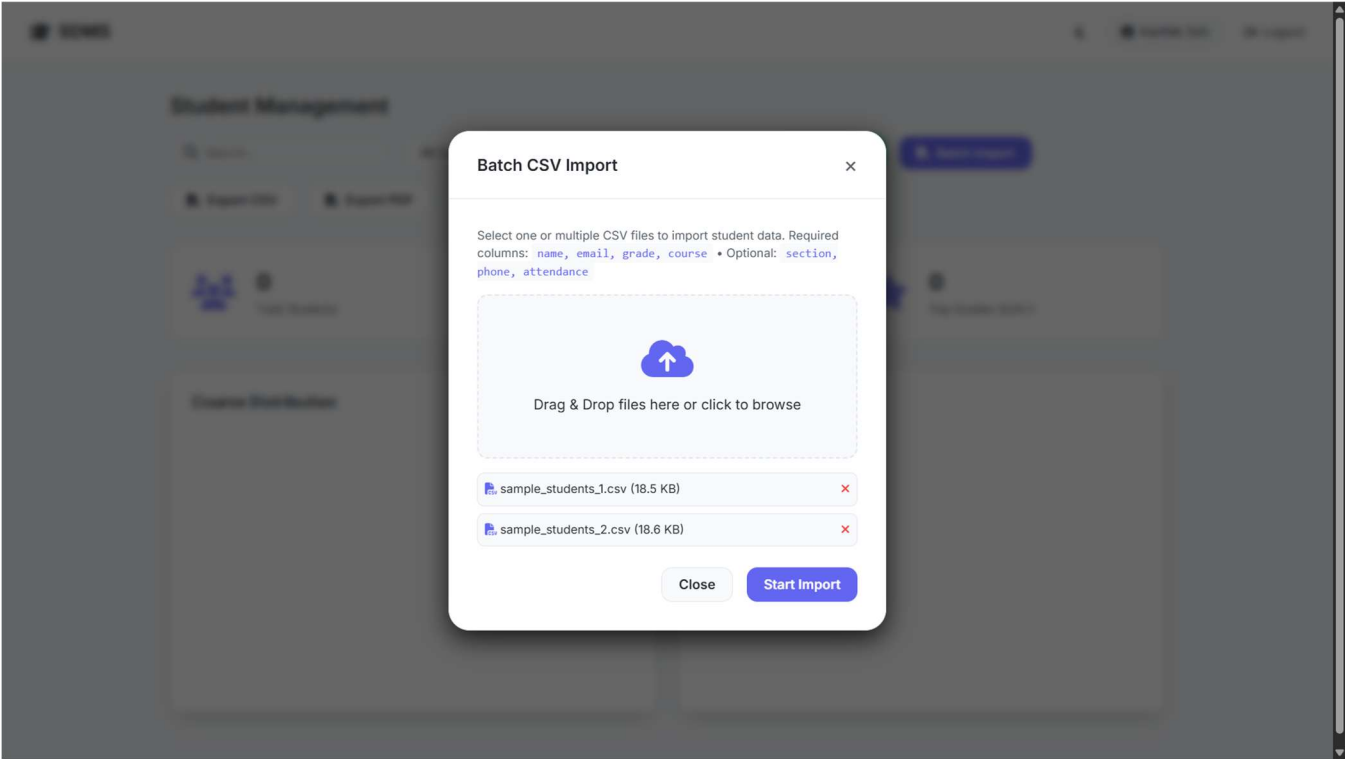
Create a password (min 6 chars)

Sample: admin123 (use at least 6 characters)

Role

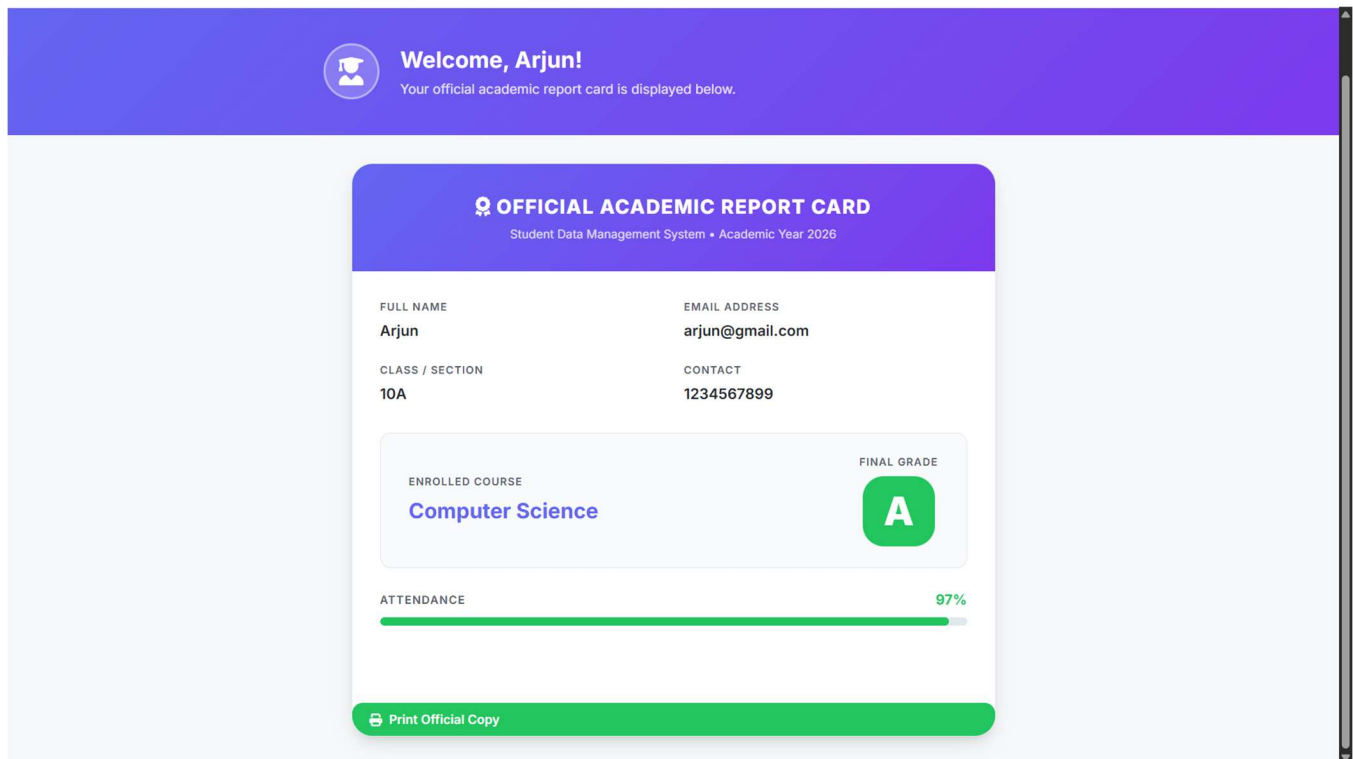
Staff / Instructor

Register



NAME	EMAIL	SECTION	GRADE	COURSE	ATTENDANCE	ACTIONS
Zara Chandra	zara.chandra.10b.55@school.edu	10B	A	Computer Science	94%	  
Yash Desai	yash.desai.10b.30@school.edu	10B	A	Computer Science	100%	  
Vikram Verma	vikram.verma.10b.65@school.edu	10B	A	Computer Science	73%	  
Vikram Desai	vikram.desai.10b.02@school.edu	10B	B	Computer Science	72%	  
Vihaan Singh	vihaan.singh.10b.05@school.edu	10B	A	Computer Science	75%	  
Ved Shah	ved.shah.10b.21@school.edu	10B	B+	Computer Science	91%	  
Uday Rao	uday.rao.10b.18@school.edu	10B	B+	Computer Science	88%	  
Trisha Menon	trisha.menon.10b.23@school.edu	10B	B	Computer Science	93%	  
Tejas Malhotra	tejas.malhotra.10b.36@school.edu	10B	B+	Computer Science	75%	  
Tejas Jain	tejas.jain.10b.42@school.edu	10B	B+	Computer Science	81%	  

sdms_students_export (1).pdf		1 / 3	100%	    	   
Student Data Report					
Generated on: 4/24/2026					
Name	Email	Grade	Course		
Zara Verma	zara.verma.10a.31@school.edu	B	Computer Science		
Zara Ghosh	zara.ghosh.10a.57@school.edu	B+	Computer Science		
Yash Menon	yash.menon.10a.69@school.edu	B+	Computer Science		
Vikram Fernandez	vikram.fernandez.10a.23@school.edu	B	Computer Science		
Vihaan Ghosh	vihaan.ghosh.10a.13@school.edu	B	Computer Science		
Ved Kulkarni	ved.kulkarni.10a.35@school.edu	A	Computer Science		
Vaishnavi Kulkarni	vaishnavi.kulkarni.10a.49@school.edu	B	Computer Science		
Vaishnavi Iyer	vaishnavi.iyer.10a.30@school.edu	A	Computer Science		
Uday Sharma	uday.sharma.10a.06@school.edu	B+	Computer Science		
Uday Malhotra	uday.malhotra.10a.37@school.edu	B	Computer Science		
Tejas Verma	tejas.verma.10a.12@school.edu	B+	Computer Science		
Tejas Kapoor	tejas.kapoor.10a.58@school.edu	B	Computer Science		
Tejas Jain	tejas.jain.10a.33@school.edu	B+	Computer Science		
Tara Chandra	tara.chandra.10a.41@school.edu	B	Computer Science		
Tanvi Rao	tanvi.rao.10a.10@school.edu	A	Computer Science		
Tanvi Menon	tanvi.menon.10a.18@school.edu	B+	Computer Science		
Tanvi Kapoor	tanvi.kapoor.10a.64@school.edu	B	Computer Science		
Tanvi Gupta	tanvi.gupta.10a.62@school.edu	B	Computer Science		
Tanvi Desai	tanvi.desai.10a.68@school.edu	B	Computer Science		
Tanvi Agarwal	tanvi.agarwal.10a.59@school.edu	B	Computer Science		
Siya Iyer	siya.iyer.10a.28@school.edu	B	Computer Science		
Siddharth Verma	siddharth.verma.10a.63@school.edu	B+	Computer Science		
Shreya Verma	shreya.verma.10a.47@school.edu	B	Computer Science		
Sunil Verma	sunil.verma.10a.70@school.edu	A	Computer Science		



Result:

Thus, the Student Data Management System project was successfully organized using proper GitHub project structure and naming conventions, and a CI/CD pipeline was implemented using Azure DevOps to automate build, testing, and deployment processes, ensuring efficient development, easy maintenance, and reliable software delivery.